

# SCREENSHOTS OF ML FUNDAMENTAL VISUALISATIONS

## 1.DATASET SAVED:

```
..  ✓ Dataset saved as student_performance.csv
```

|    | Study_Hours | Attendance | Assignments | Marks |
|----|-------------|------------|-------------|-------|
| 0  | 1           | 60         | 2           | 35    |
| 1  | 2           | 65         | 3           | 40    |
| 2  | 3           | 70         | 4           | 45    |
| 3  | 4           | 72         | 5           | 50    |
| 4  | 5           | 75         | 5           | 58    |
| 5  | 6           | 80         | 6           | 65    |
| 6  | 7           | 82         | 7           | 72    |
| 7  | 8           | 85         | 7           | 78    |
| 8  | 9           | 88         | 8           | 82    |
| 9  | 10          | 90         | 8           | 88    |
| 10 | 11          | 92         | 9           | 92    |
| 11 | 12          | 95         | 10          | 96    |

## 2.DATASET EXPLORATION :

## PART 1 – DATASET EXPLORATION

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### ✚ First 5 Rows:

|   | Study_Hours | Attendance | Assignments | Marks |
|---|-------------|------------|-------------|-------|
| 0 | 1           | 60         | 2           | 35    |
| 1 | 2           | 65         | 3           | 40    |
| 2 | 3           | 70         | 4           | 45    |
| 3 | 4           | 72         | 5           | 50    |
| 4 | 5           | 75         | 5           | 58    |

### ✚ Last 5 Rows:

|    | Study_Hours | Attendance | Assignments | Marks |
|----|-------------|------------|-------------|-------|
| 7  | 8           | 85         | 7           | 78    |
| 8  | 9           | 88         | 8           | 82    |
| 9  | 10          | 90         | 8           | 88    |
| 10 | 11          | 92         | 9           | 92    |
| 11 | 12          | 95         | 10          | 96    |

### ✚ Dataset Shape (Rows x Columns):

(12, 4)

### ✚ Column Names:

['Study\_Hours', 'Attendance', 'Assignments', 'Marks']

### ✚ Data Types:

Study\_Hours     int64  
Attendance     int64  
Assignments     int64  
Marks           int64  
dtype: object

### ✚ Summary Statistics:

|       | Study_Hours | Attendance | Assignments | Marks     |
|-------|-------------|------------|-------------|-----------|
| count | 12.000000   | 12.000000  | 12.000000   | 12.000000 |
| mean  | 6.500000    | 79.500000  | 6.166667    | 66.750000 |
| std   | 3.605551    | 11.172206  | 2.443296    | 21.080474 |
| min   | 1.000000    | 60.000000  | 2.000000    | 35.000000 |
| 25%   | 3.750000    | 71.500000  | 4.750000    | 48.750000 |
| 50%   | 6.500000    | 81.000000  | 6.500000    | 68.500000 |
| 75%   | 9.250000    | 88.500000  | 8.000000    | 83.500000 |
| max   | 12.000000   | 95.000000  | 10.000000   | 96.000000 |

## 3.FEATURES AND LABELS :

## PART 2 — FEATURES AND LABEL

=====

### 📌 Features (X):

|    | Study_Hours | Attendance | Assignments |
|----|-------------|------------|-------------|
| 0  | 1           | 60         | 2           |
| 1  | 2           | 65         | 3           |
| 2  | 3           | 70         | 4           |
| 3  | 4           | 72         | 5           |
| 4  | 5           | 75         | 5           |
| 5  | 6           | 80         | 6           |
| 6  | 7           | 82         | 7           |
| 7  | 8           | 85         | 7           |
| 8  | 9           | 88         | 8           |
| 9  | 10          | 90         | 8           |
| 10 | 11          | 92         | 9           |
| 11 | 12          | 95         | 10          |

### 📌 Label (y):

|    |    |
|----|----|
| 0  | 35 |
| 1  | 40 |
| 2  | 45 |
| 3  | 50 |
| 4  | 58 |
| 5  | 65 |
| 6  | 72 |
| 7  | 78 |
| 8  | 82 |
| 9  | 88 |
| 10 | 92 |
| 11 | 96 |

Name: Marks, dtype: int64

Feature columns: ['Study\_Hours', 'Attendance', 'Assignments']  
Label column : Marks

## 4. TRAIN-TEST SPLIT :

```
.. =====  
PART 4 — TRAIN-TEST SPLIT  
=====
```

Total Samples : 12  
Training (80%) : 9  
Testing (20%) : 3

X\_train shape: (9, 3)  
X\_test shape: (3, 3)  
y\_train shape: (9,)  
y\_test shape: (3,)

## 5. TRAIN LINEAR REGRESSION :

```
=====
PART 5 – TRAIN LINEAR REGRESSION
=====
```

✅ Model Trained!

▴ Coefficients (Slope):

Study\_Hours: 3.4549

Attendance: 1.3449

Assignments: -2.4027

▴ Intercept: -48.1006

Answers:

1. Slope → Coefficients of each feature.  
Shows how much Marks change per unit increase.
2. Intercept → Value of Marks when all features are 0.
3. Meaning → Higher Study\_Hours, Attendance & Assignments  
→ Higher predicted Marks.

## 6.PREDICTIONS :

```
=====
PART 6 – PREDICTIONS ON NEW DATA
=====
```

New Student Data with Predictions:

| Study_Hours | Attendance | Assignments | Predicted_Marks |
|-------------|------------|-------------|-----------------|
| 5           | 80         | 5           | 64.75           |
| 8           | 90         | 8           | 81.36           |
| 12          | 95         | 10          | 97.10           |

## 7.ACTUAL VS PREDICTED :

```
=====
PART 7 – ACTUAL VS PREDICTED
=====
```

|           | Actual Marks | Predicted Marks | Difference |
|-----------|--------------|-----------------|------------|
| Student 1 | 92           | 92.01           | -0.01      |
| Student 2 | 88           | 88.27           | -0.27      |
| Student 3 | 35           | 31.24           | 3.76       |

```
! Highest Prediction Error: Student 3
Error: 3.76 marks
```

## 8.MODEL EVALUATION :

```
*** =====
PART 8 – MODEL EVALUATION
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```

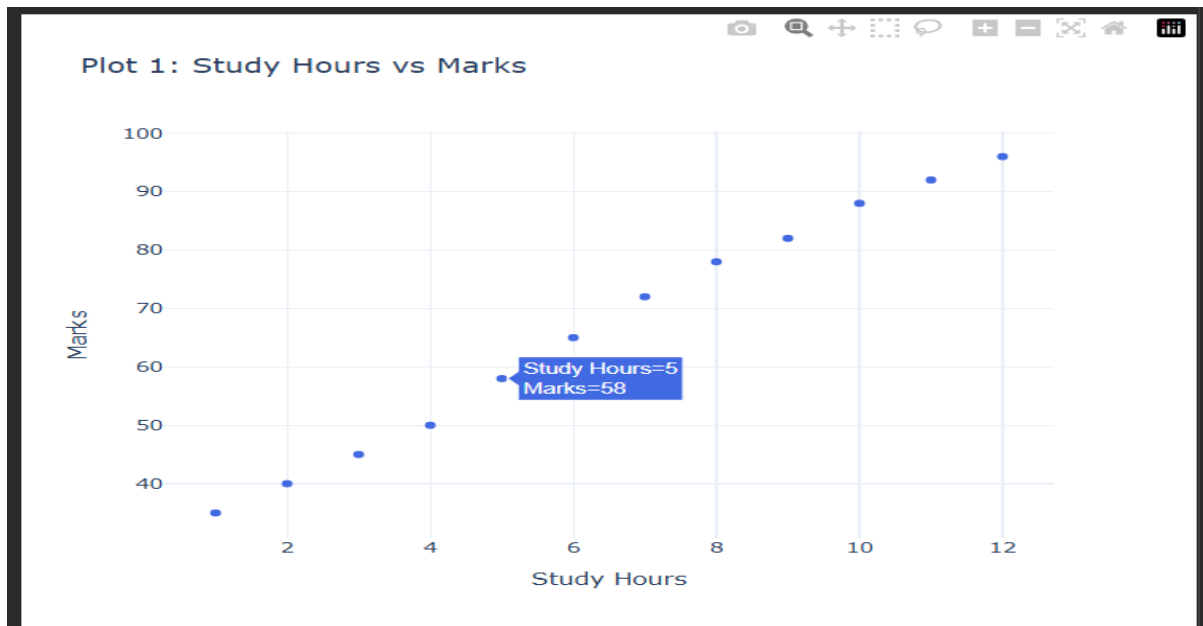
```
MAE (Mean Absolute Error) : 1.35
R2 Score                  : 0.9930
```

Answers:

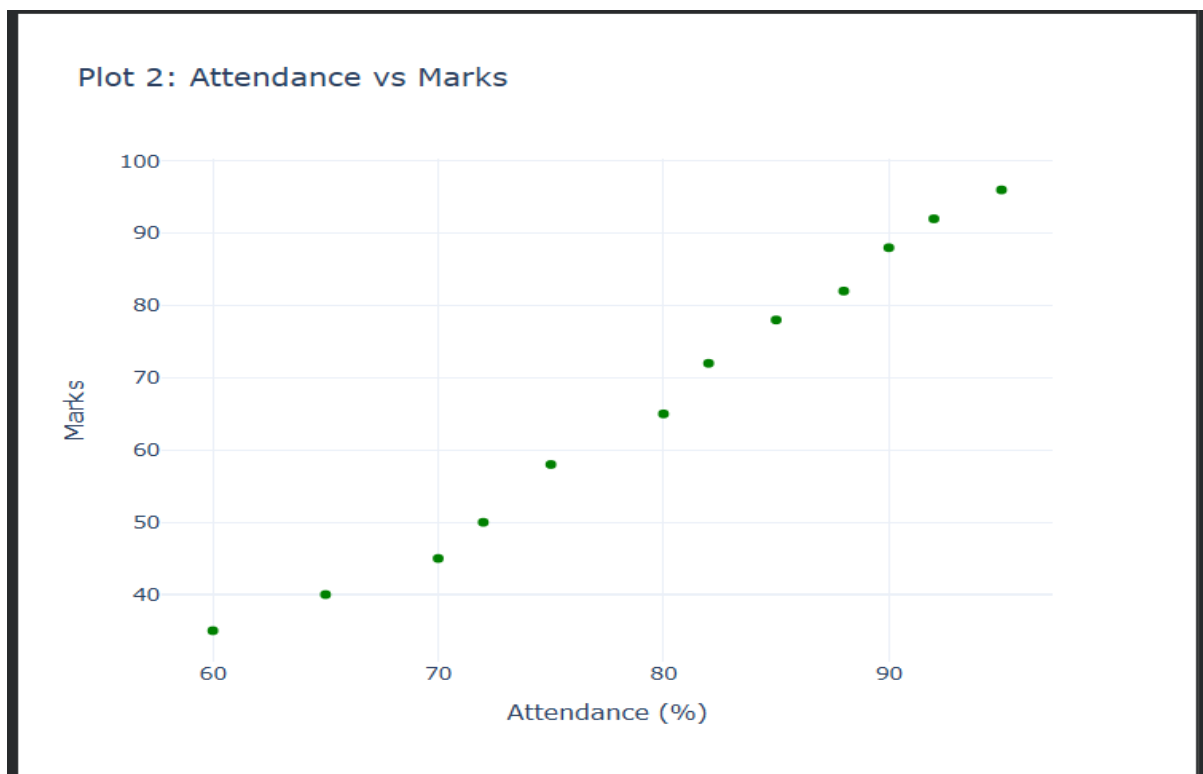
1. MAE → Average absolute difference between actual and predicted marks. Lower is better.
2. R<sup>2</sup> Score → How well the model explains the data.  
Ranges 0 to 1. Closer to 1 = better model.
3. Performance → If R<sup>2</sup> is above 0.90, the model is performing excellently!

## 9.VISUALISATIONS :

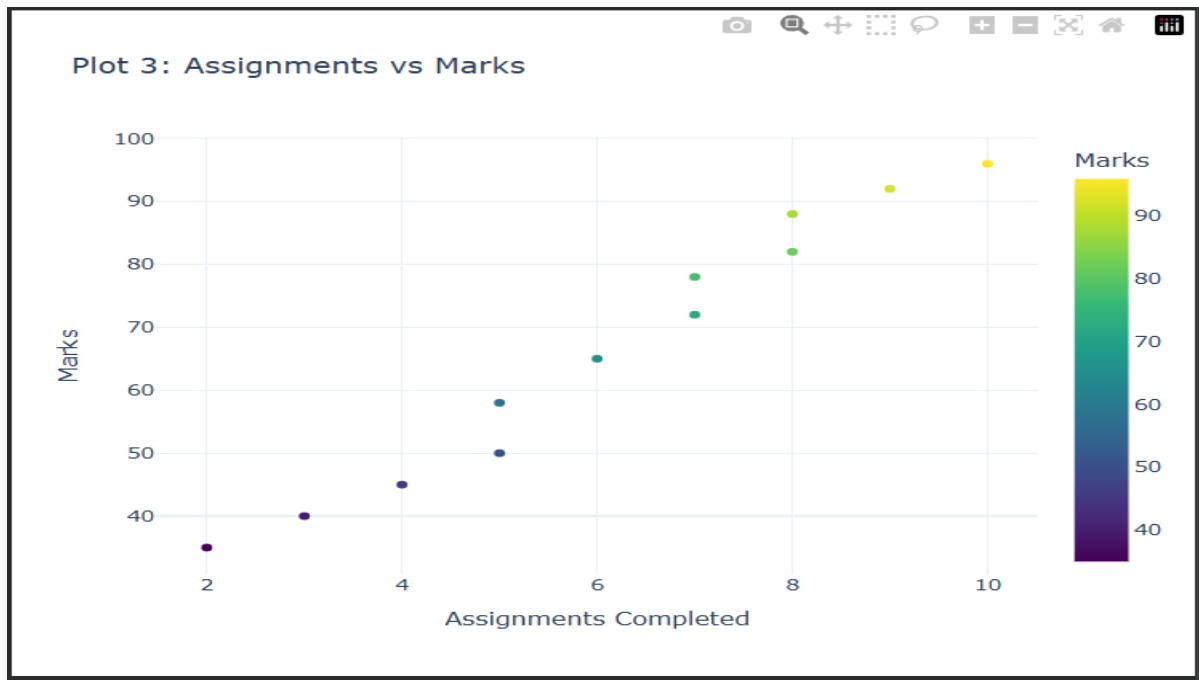
### 9A. PLOT-1:STUDY HOURS VS MARKS :



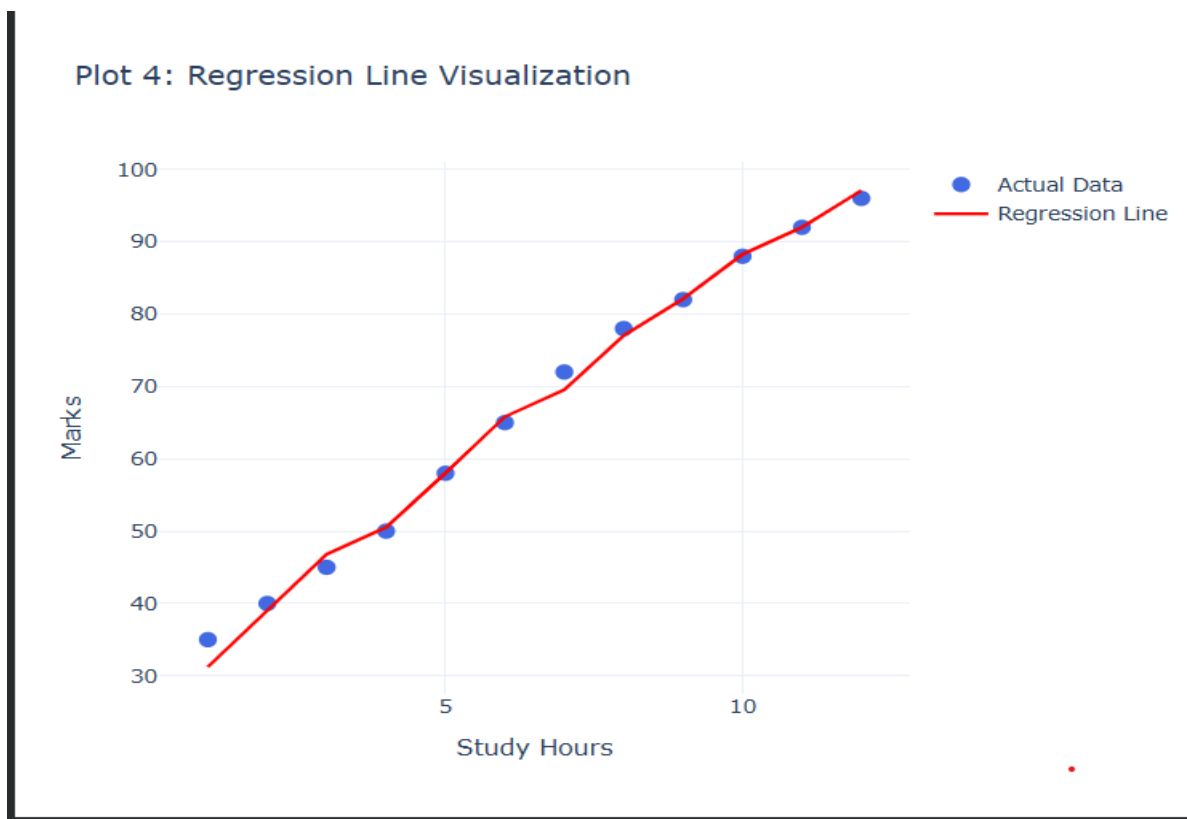
## 9B.PLOT-2:ATTENDANCE VS MARKS :



## 9C . PLOT-3- ASSIGNMENTS VS MARKS :



## 9D.PLOT-4 -REGRESSION LINES :



## 9E .PLOT-5-PREDICTED MARKS TREND(STUDY HOURS 1-15)

Plot 5: Predicted Marks Trend (Study Hours 1–15)

